

EXECUTIVE SUMMARY

Predictive Modeling for Course Demand and Revenue Forecasting on EduPro Course Demand, Revenue Forecasting & Predictive Intelligence Analysis

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Program: Unified Mentor Internship
Year: 2026

1. Executive Overview

This executive summary presents the key findings from the Predictive Modeling for Course Demand and Revenue Forecasting project on EduPro. The project uses historical course, teacher, user, and transaction data to develop predictive intelligence for understanding course enrollment demand and revenue performance. The analysis uses Python, Pandas, NumPy, Scikit-learn, and Streamlit to transform historical data into predictive insights and an interactive dashboard. The findings are intended to support data-driven decisions related to course planning, pricing, instructor allocation, and future course performance.

2. Project Objective

The primary objective of the project is to develop a predictive analytics solution that can estimate future course demand and revenue using historical EduPro data. The project focuses on predicting enrollment count, analyzing course-level revenue, and comparing revenue performance across course categories. It also aims to identify important factors influencing course demand and provide an interactive platform for exploring predictive results.

3. Analytical Approach

The analysis followed a structured data science workflow. The dataset was first examined and prepared by combining course, teacher, user, and transaction information. Transaction data was aggregated at the course level to calculate enrollment and revenue-related measures. Feature engineering was then performed to create variables such as revenue per enrollment, revenue efficiency, price groups, duration groups, rating groups, teacher quality score, teacher experience, and teacher rating. Machine learning models were developed and evaluated using appropriate performance metrics, followed by feature importance analysis and Streamlit dashboard development.

Dataset → Data Preparation → Feature Engineering → Exploratory Analysis → Model Development → Model Evaluation → Predictive Dashboard

4. Key Findings

✓ Course Demand

Historical course enrollment was analyzed to identify factors associated with course demand. Course characteristics such as rating, duration, instructor experience, teacher count, and instructor rating were among the important variables considered by the Random Forest model. This indicates that both course-level and instructor-level characteristics can contribute to understanding enrollment demand.

✓ Revenue Performance

Course revenue was analyzed using transaction-level data aggregated at the course level. Revenue varied considerably across courses, with a large number of transactions having zero monetary value. This highlights the importance of distinguishing enrollment activity from direct revenue generation when evaluating course performance.

✓ Instructor Influence

Instructor-related characteristics were incorporated into the predictive dataset through measures such as average teacher experience, average teacher rating, teacher count, and teacher quality score. These features provide additional information for understanding how instructor characteristics may relate to course performance.

✓ Model Development

A Random Forest Regression model was developed for enrollment prediction. The model used course, instructor, and historical performance features to estimate enrollment demand. Cross-validation and hyperparameter tuning were performed to evaluate and improve the model configuration.

5. Key Performance Indicators

KPI	Result
Dataset Size	10,000 Transactions
Courses Analyzed	60
Users Analyzed	3,000
Teachers Analyzed	60
Average Transaction Amount	91.13
Maximum Transaction Amount	490.90
Zero-Amount Transactions	64.03%

6. Business Implications

The findings demonstrate how predictive analytics can support EduPro in moving from historical reporting toward more proactive course planning. Course ratings, duration, instructor experience, and instructor ratings can be considered when evaluating potential course demand. Revenue analysis can help identify courses and categories that require greater attention from a pricing and content-planning perspective. The combination of predictive modeling and interactive visualization also allows stakeholders to examine course performance more efficiently and make decisions based on quantitative evidence.

7. Recommendations

- **Use Predictive Demand Analysis**

Use enrollment predictions to identify courses that may have stronger or weaker future demand and support course planning decisions.

- **Consider Instructor Quality**

Include instructor experience and ratings when evaluating potential course performance and instructor allocation.

- **Optimize Course Pricing**

Analyze the relationship between course price, enrollment, and revenue to support more informed pricing decisions.

- **Monitor Revenue by Category**

Compare category-level revenue regularly to identify stronger-performing and underperforming areas.

- **Continue Model Evaluation**

Regularly evaluate prediction performance using MAE, RMSE, and R^2 and retrain models when new historical data becomes available.

- **Use the Streamlit Dashboard**

Use the interactive dashboard to provide stakeholders with accessible course demand, revenue, category, and feature-importance insights.

8. Limitations

The findings are based on the available EduPro dataset and should not be interpreted as representing the performance of all online learning platforms. The dataset contains 60 courses, 60 teachers, 3,000 users, and 10,000 transactions, which limits the generalization of the findings. A substantial proportion of transactions have zero monetary value, which can affect revenue-related analysis. The predictive models identify relationships within the available historical data but cannot guarantee future course performance. Model performance may also be affected by the limited number of course-level observations available for training and evaluation.

9. Conclusion

Overall, the EduPro Predictive Modeling project provides a data-driven framework for understanding course demand and revenue performance. By combining historical transaction data with course and instructor characteristics, the project demonstrates how machine learning can be used to support future-oriented decision-making. The developed predictive models, feature engineering techniques, and Streamlit dashboard provide a foundation for improving course planning, pricing analysis, instructor allocation, and category-level revenue monitoring. The project therefore helps transform EduPro's historical data into actionable predictive intelligence.